**Business Economics- TCS 2112**

**Government Intervention**

**Tutorial 04- Part I**

**01. The demand and supply equations for a competitive Vegetable market are shown below.**

Qd= 720-4P

Qs= -120+2P

Required:

i. Calculate the equilibrium price and quantity.
ii. Draw the diagram based on the above information.
iii. Calculate the consumer and producer surplus at the equilibrium.
iv. Calculate the total surplus at the equilibrium.

**02. The demand and supply equations for a competitive rice market are shown below.**

Qd= 1320-6P

Qs= -680+4P

Required:

i. Calculate the equilibrium price and quantity.
ii. Draw the diagram based on the above information.
iii. Calculate the consumer and producer surplus at the equilibrium.
iv. Calculate the total surplus at the equilibrium.
v. If the government legally sets a maximum price of Rs. 180 per kilogram for rice,
    a. will it create an excess demand or an excess supply? Calculate the amount of the shortage or surplus.
    b. Calculate the consumer and producer surplus after maximum price
    c. Calculate the black-market price.

d. Calculate the consumer surplus after the black-market price
e. Calculate the dead weight lost.

**03. The demand and supply equations for a competitive dairy market are shown below.**

Qd= 180 – 2P

Qs= -15+P

Required;

i. Calculate the equilibrium price and quantity.
ii. Draw the diagram based on the above information.
iii. Calculate the consumer and producer surplus at the equilibrium.
iv. Calculate the total surplus at the equilibrium.
v. If the government has decided to introduce a minimum price of Rs. 72 per liter of milk;
    a. What is the excess supply in the market
    b. Calculate the consumer and producer surplus after minimum price
    c. Calculate the amount of dead weight loss?
vi. If government decided to buy the excess supply,
    a. Calculate the government expenditure
    b. Calculate the producer surplus
    c. Deadweight loss